Natural sonni to'g'ri kasrga bo'lish

5 soatda nechta chorak soat bor? 10 m matodan $\frac{5}{6}$ m dan nechta bo'lak chiqadi?

1-QISM

Mavzu

ESLAB OLAYLIK

- $18 : 6 = 3$ — 18 ichida 6 uch marta bor.
- Natural sonni kasr ko'rinishida yozish mumkin: $5 = \frac{5}{1}$.
- $\frac{17}{5} = 3\frac{2}{5}$ — 17 ichida 5 uch marta bor, 2 qoladi.



Bitta nonni oltiga bo'lsam, olti bo'lak. Uchta nonni-chi?

USUL · BUTUNLARNI BO'LAKLARGA BO'LISH

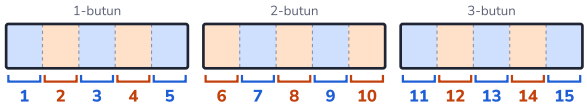
1 butunda $\frac{1}{6}$ dan 6 ta bor: $1 : \frac{1}{6} = 6$.

1 butun → 6 ta bo'lak 3 butun → $3 \cdot 6 = 18$ ta bo'lak

Demak, $3 : \frac{1}{6} = 18$.

1-MISOL · YO'L

3 km yo'l $\frac{1}{5}$ km li teng qismlarga bo'lindi. Nechta qism chiqdi?



1. 1 km da 5 ta qism.
2. 3 km da: $3 : \frac{1}{5} = 3 \cdot 5 = 15$ ta qism.

2-MISOL · SOAT

5 soatda nechta chorak soat ($\frac{1}{4}$ soat) bor?

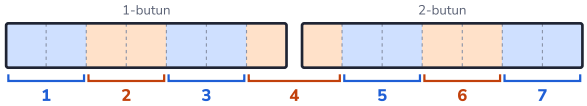
1. 1 soatda 4 ta chorak: $5 : \frac{1}{4} = 5 \cdot 4 = 20$ ta.
2. Tekshiramiz: chorak soat — 15 minut, $20 \cdot 15 = 300$ minut, ya'ni 5 soat.

USUL · BO'LAKLARNI GURUHLASH

Bo'luvchi $\frac{2}{7}$ bo'lsa, $\frac{1}{7}$ bo'laklarni 2 tadan guruhlab sanaymiz.

1-MISOL · TO'LIQ GURUHLAR

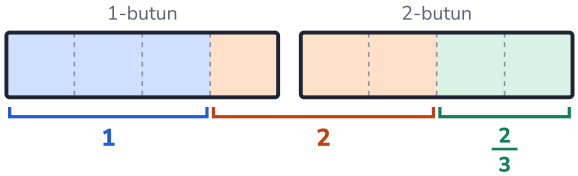
2 ichida $\frac{2}{7}$ necha marta bor?



1. 2 butunda $2 \cdot 7 = 14$ ta $\frac{1}{7}$ bo'lak.
2. 2 tadan guruhlaymiz: $2 : \frac{2}{7} = 14 : 2 = 7$.

2-MISOL · CHALA GURUH

$2 : \frac{3}{4}$ ni toping.



1. 2 butunda $2 \cdot 4 = 8$ ta chorak.
2. 3 tadan guruhlaymiz: 2 ta to'liq guruh, yana 2 ta bo'lak ortdi — bu guruhning $\frac{2}{3}$ qismi.
3. 8 ta bo'lak 3 tadan: $2 : \frac{3}{4} = 8 : 3 = \frac{8}{3} = 2\frac{2}{3}$.

100 ni $\frac{7}{9}$ ga bo'lishda ham rasm chizamanmi?



Bo'lish	Ko'paytirish
$3 : \frac{1}{5} = 15$	$3 \cdot \frac{5}{1} = 15$
$2 : \frac{2}{7} = 7$	$2 \cdot \frac{7}{2} = 7$
$2 : \frac{3}{4} = 2\frac{2}{3}$	$2 \cdot \frac{4}{3} = 2\frac{2}{3}$

Qarang: ikkinchi ustunda kasr ag'darilgan. Ag'darilgan kasr — **teskari kasr**: $\frac{3}{4}$ ning teskarisi $\frac{4}{3}$.

QOIDA

Natural sonni kasrga bo'lish:

$$4 : \frac{2}{9} = 4 \cdot \frac{9}{2} = 18$$

- 4 — o'z holida qoldiring;
- $:$ ni $\cdot$ bilan almashtiring;
- $\frac{2}{9}$ ning teskarisini yozing: $\frac{9}{2}$.

1-MISOL · JAVOB BUTUN SON

$16 : \frac{4}{5}$ ni hisoblang.

1. $16 : \frac{4}{5} = 16 \cdot \frac{5}{4} = \frac{16 \cdot 5}{4}$.
2. 16 va 4 ni 4 ga qisqartiramiz: $4 \cdot 5 = 20$.

2-MISOL · JAVOB ARALASH SON

$7 : \frac{3}{5}$ ni hisoblang.

1. $7 : \frac{3}{5} = 7 \cdot \frac{5}{3} = \frac{35}{3}$.
2. 35 : 3 = 11, qoldiq 2: $\frac{35}{3} = 11\frac{2}{3}$.

2-QISM

Misollar ishlash

1-MISOL · QISQARTIRIB HISOBLASH

$10 : \frac{4}{7}$ ni hisoblang.

1. $10 : \frac{4}{7} = 10 \cdot \frac{7}{4}$.
2. 10 va 4 ni 2 ga qisqartiramiz: $\frac{5 \cdot 7}{2} = \frac{35}{2} = 17\frac{1}{2}$.

2-MISOL · AYLANMA YO'LAK

Bog'dagi aylanma yo'lakning uzunligi $\frac{4}{5}$ km. Bobur 6 km yugurdi. U nechta to'liq aylana yugurdi? Oxirgi, chala aylanada necha km yugurdi?

1. $6 : \frac{4}{5} = 6 \cdot \frac{5}{4} = \frac{30}{4} = 7\frac{1}{2}$.
2. To'liq aylanalar — butun qism: 7 ta.
3. 7 aylana: $7 \cdot \frac{4}{5} = \frac{28}{5} = 5\frac{3}{5}$ km.
4. Chala aylanada: $6 - 5\frac{3}{5} = \frac{2}{5}$ km.

3-MISOL · TAXTA

Duradgor uzunligi 3 m bo'lgan taxtani har biri $\frac{3}{8}$ m li bo'laklarga kesdi. U necha marta kesgan?

1. Bo'laklar soni: $3 : \frac{3}{8} = 3 \cdot \frac{8}{3} = 8$ ta.
2. Taxta 2 bo'lakka 1 marta, 3 bo'lakka 2 marta kesiladi: kesishlar bo'laklardan bitta kam.
3. $8 - 1 = 7$ marta.

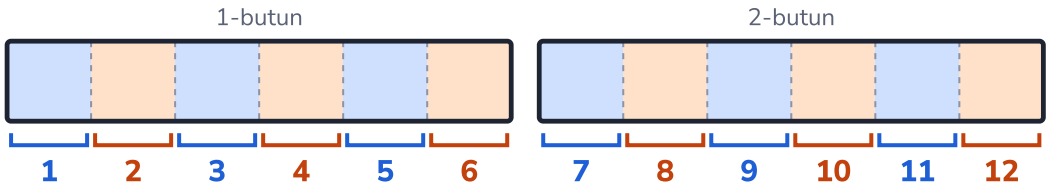
DIQQAT!

1. Teskarisi yozilmagan:
   ✗ $8 : \frac{2}{3} = 8 \cdot \frac{2}{3} = 5\frac{1}{3}$ → ✓ $8 \cdot \frac{3}{2} = 12$
   $12 > 8$ — to'g'ri kasrga bo'lsak, son kattalashadi.
2. Son ag'darilgan:
   ✗ $6 : \frac{1}{5} = \frac{1}{6} \cdot \frac{1}{5}$ → ✓ $6 \cdot \frac{5}{1} = 30$
3. Suratga bo'linmagan:
   ✗ $4 : \frac{3}{7} = 4 \cdot 7 = 28$ → ✓ $4 \cdot \frac{7}{3} = \frac{28}{3} = 9\frac{1}{3}$
4. «Nechta to'liq?» savoli:
   ✗ $6\frac{2}{3}$ ta quti → ✓ **6 ta quti**
5. 8 bo'lakka kesish:
   ✗ 8 marta kesiladi → ✓ **7 marta kesiladi**

3-QISM

Mustahkamlash

1. Rasm qaysi bo'lishni ko'rsatadi? Javobi nechaga teng?



2. Bo'lishni bajaring: a) $5 : \frac{1}{3}$; b) $7 : \frac{1}{4}$; c) $9 : \frac{1}{2}$; d) $4 : \frac{1}{10}$.
3. Nuqtalar o'rniga sonni yozing: a) 1 butunda ... ta $\frac{1}{8}$ bor; b) 6 soatda ... ta yarim soat bor; c) 3 m da ... ta $\frac{1}{10}$ m bor.
4. Kasrning teskarisini yozing: $\frac{3}{8}$; $\frac{1}{6}$; $\frac{9}{4}$; $\frac{5}{12}$.
5. Teskari kasrga ko'paytirib toping: a) $6 : \frac{3}{5}$; b) $8 : \frac{4}{9}$; c) $14 : \frac{7}{10}$; d) $12 : \frac{4}{7}$.
6. Javobni aralash son ko'rinishida yozing: a) $5 : \frac{2}{9}$; b) $2 : \frac{3}{7}$; c) $3 : \frac{4}{7}$; d) $11 : \frac{5}{6}$.
7. Taqqoslang ($>$, $<$ yoki $=$): a) $4 : \frac{4}{9}$ va $4 \cdot \frac{9}{4}$; b) $7 : \frac{7}{9}$ va 7; c) $2 : \frac{1}{9}$ va $3 : \frac{1}{6}$.
8. Nodira shunday yechdi: $12 : \frac{3}{8} = 12 \cdot \frac{3}{8} = 4\frac{1}{2}$. Xato qayerda? To'g'ri javob nechaga teng?
9. Ifodaning qiymatini toping: a) $6 : \frac{3}{4} + 2$; b) $10 - 4 : \frac{4}{7}$.
10. Katakka qaysi son yoziladi? a) $\square : \frac{1}{4} = 20$; b) $3 : \frac{1}{\square} = 21$; c) $5 : \frac{\square}{7} = 7$.
11. Qaysi biri katta: $8 : \frac{4}{5}$ yoki $7 : \frac{7}{12}$? Nechaga katta?

Matnli masalalar

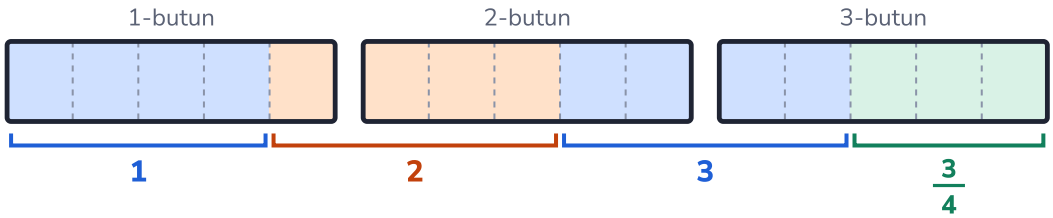
1. Bitta yostiq jildini tikishga $\frac{5}{6}$ m mato ketadi. 10 m matodan nechta jild tikish mumkin?

2. Robot bitta o'yinchoqni $\frac{2}{5}$ minutda yig'adi. 8 minutda u nechta o'yinchoq yig'adi?
3. Bo'yoqchida 7 kg bo'yoq bor. Bitta eshikni bo'yashga $\frac{3}{4}$ kg bo'yoq ketadi. U nechta eshikni to'liq bo'yay oladi va qancha bo'yoq ortib qoladi?
4. Usta uzunligi 6 m bo'lgan quvurni har biri $\frac{3}{5}$ m li bo'laklarga kesdi. U necha marta kesgan?
5. Qandolatchida 5 kg un bor. Bitta tortga $\frac{5}{6}$ kg un ketadi. Har bir tort 8 bo'lakka kesiladi. Jami nechta bo'lak tort chiqadi?

4-QISM

Mustaqil ish

1. Hisoblang: a) $8 : \frac{1}{7}$; b) $15 : \frac{5}{9}$; c) $5 : \frac{4}{9}$.
2. Rasmdagi bo'lishni yozing va javobini toping.



3. Uzunligi 15 m bo'lgan yo'lakka uzunligi $\frac{3}{4}$ m bo'lgan plitkalar bir qator qilib uchma-uch terildi. Nechta plitka ketgan?
4. Sartarosh bitta mijozga $\frac{5}{6}$ soat sarflaydi. 8 soatlik ish kunida u ko'pi bilan nechta mijozga xizmat qiladi? Qancha vaqti ortib qoladi?
5. Uzunligi 4 m bo'lgan gul jo'yagi bo'ylab har $\frac{2}{7}$ m da bittadan gul ekildi. Jo'yakning boshida ham, oxirida ham gul bor. Nechta gul ekilgan? *(Avval oraliqlar sonini toping. Har oraliqning boshida bitta gul bor, jo'yak oxirida yana bittasi.)*
6. Qaysi sonni $\frac{3}{5}$ ga bo'lsak, 15 chiqadi? *(15 ta $\frac{3}{5}$ qancha bo'ladi?)*

5-QISM

Javoblar

Mustahkamlash

1-masala

$2 : \frac{1}{6} = 2 \cdot 6 = 12$ — 2 butunda 12 ta $\frac{1}{6}$ bo'lak.

2-masala

a) $5 \cdot 3 = 15$; b) $7 \cdot 4 = 28$; c) $9 \cdot 2 = 18$; d) $4 \cdot 10 = 40$.

3-masala

a) 8; b) $6 \cdot 2 = 12$; c) $3 \cdot 10 = 30$.

4-masala

$\frac{8}{3}; \frac{6}{1} = 6; \frac{4}{9}; \frac{12}{5}$.

5-masala

a) $6 \cdot \frac{5}{3} = 10$; b) $8 \cdot \frac{9}{4} = 18$; c) $14 \cdot \frac{10}{7} = 20$; d) $12 \cdot \frac{7}{4} = 21$.

6-masala

a) $5 \cdot \frac{9}{2} = \frac{45}{2} = 22\frac{1}{2}$; b) $2 \cdot \frac{7}{3} = \frac{14}{3} = 4\frac{2}{3}$; c) $3 \cdot \frac{7}{4} = \frac{21}{4} = 5\frac{1}{4}$; d) $11 \cdot \frac{6}{5} = \frac{66}{5} = 13\frac{1}{5}$.

7-masala

a) $9 = 9$; b) $9 > 7$; c) $18 = 18$.

8-masala

Kasrning teskarisi yozilmagan. To'g'ri: $12 \cdot \frac{8}{3} = 32$.

9-masala

a) $6 \cdot \frac{4}{3} = 8, 8 + 2 = 10$; b) $4 \cdot \frac{7}{4} = 7, 10 - 7 = 3$.

10-masala

a) $\square \cdot 4 = 20, \square = 5$; b) $3 \cdot \square = 21, \square = 7$; c) 5 butunda 35 ta $\frac{1}{7}$ bo'lak, guruhlar 7 ta — har guruhda 5 ta bo'lak: $\square = 5$.

11-masala

$8 \cdot \frac{5}{4} = 10, 7 \cdot \frac{12}{7} = 12$. $7 : \frac{7}{12}$ katta, 2 ga.

Matnli masalalar

1-masala

$10 : \frac{5}{6} = 10 \cdot \frac{6}{5} = 12$ ta jild.

2-masala

$8 : \frac{2}{5} = 8 \cdot \frac{5}{2} = 20$ ta o'yinchoq.

3-masala

$7 : \frac{3}{4} = \frac{28}{3} = 9\frac{1}{3}$ — 9 ta eshik. Ketgan bo'yoq: $9 \cdot \frac{3}{4} = 6\frac{3}{4}$ kg. Ortdi: $7 - 6\frac{3}{4} = \frac{1}{4}$ kg.

4-masala

Bo'laklar: $6 : \frac{3}{5} = 6 \cdot \frac{5}{3} = 10$ ta. Kesishlar: $10 - 1 = 9$ marta.

5-masala

Tortlar: $5 : \frac{5}{6} = 5 \cdot \frac{6}{5} = 6$ ta. Bo'laklar: $6 \cdot 8 = 48$ ta.

Mustaqil ish

1-masala

a) $8 \cdot 7 = 56$; b) $15 \cdot \frac{9}{5} = 27$; c) $5 \cdot \frac{9}{4} = \frac{45}{4} = 11\frac{1}{4}$.

2-masala

$3 : \frac{4}{5}$: 15 ta bo'lakdan 3 ta to'liq guruh va $\frac{3}{4}$ guruh. $3 \cdot \frac{5}{4} = \frac{15}{4} = 3\frac{3}{4}$.

3-masala

$15 : \frac{3}{4} = 15 \cdot \frac{4}{3} = 20$ ta plitka.

4-masala

$8 : \frac{5}{6} = \frac{48}{5} = 9\frac{3}{5}$ — 9 ta mijoz. Sarflangan vaqt: $9 \cdot \frac{5}{6} = 7\frac{1}{2}$ soat. Ortdi: $8 - 7\frac{1}{2} = \frac{1}{2}$ soat.

5-masala

Oraliqlar: $4 : \frac{2}{7} = 4 \cdot \frac{7}{2} = 14$ ta. Gullar: $14 + 1 = 15$ ta.

6-masala

$15 \cdot \frac{3}{5} = 9$. Tekshiramiz: $9 : \frac{3}{5} = 9 \cdot \frac{5}{3} = 15$.